

Organic chemistry - Some basic principles &

Technique

1. General Organic Chemistry :-

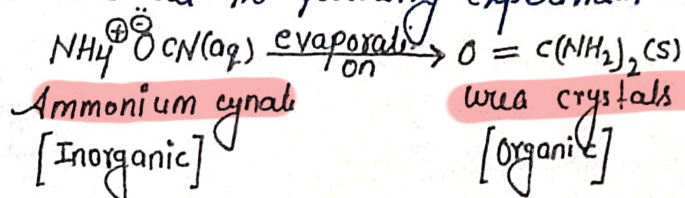
Introduction :- Berzelius coined the term "Organic chemistry" in 1807 to describe the study of compound derived from natural sources. This was based on the vitalism theory, which stated that all living system possessed a "vital force" that non-living system lacked. Organic compounds were thought to be fundamentally different from inorganic compounds because they were derived from living natural sources.

vital force
Living system

Non-vital force
Non-living system.

Philosophically the vital force could be thought of as the mysterious force that God instilled in living system.

Friedrich Wohler :- became Berzelius student in 1823. Wohler made a discovery in 1828 that changed the definition of organic chemistry. Wohler carried out the following experiment.



Wohler was able to successfully synthesize an inorganic compound. Many others followed suit, synthesizing organic compounds from inorganic compounds. As a result, the theory of vitalism and the definition of organic

chemistry became meaningless.

The presence of carbon however was shared by all of the compounds synthesized above. Catenation is a unique property of carbon. Carbon can form long chains & rings with other carbon atoms [self-catenation] as well as with atoms of many other elements in the periodic table [cross catenation]. As a result, carbon can form a wide range of compounds.

The vital force is probably explained by the fact that most of the living life & life-sustaining functions are performed by carbon atoms/compounds.

Eg :- Human tissue & skin are formed by proteins; respiration is possible because of haemoglobin; information in our genes is carried out in the form of DNA/RNA and so on.

∴ The detailed study of fundamental concepts and factors that govern the progression and outcome of reactions is known as general organic chemistry.

NOTE :-

Before transforming into product, bonds are typically formed and broken in a series of discrete steps. The mechanism of the reaction is a detailed sequential description of all the steps.

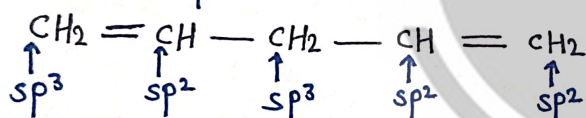
Size of Hybrid Orbitals :- The size of hybrid orbital decrease as the % of s character increase. As a result the size of the hybrid orbital is $sp^3 > sp^2 > sp$

Electronegativity of Hybrid Orbital :-

As the % of s character increase. so does the electronegativity of the hybrid orbital As a result the EN of Hybrid orbital is $sp > sp^2 > sp^3$

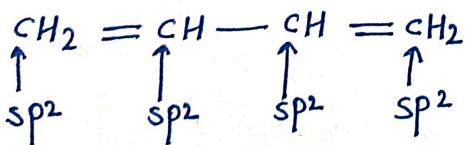
Dienes :- Dienes are organic compound that have two double bond in them. dienes are classified into three types:
1. Isolated 2. Conjugated 3. cumulated.

1. Isolated dienes :- In this case double bond are separated by atleast one sp^3 carbon.



molecule	σ	π	L.P	hybridization	shape
CH_4	4	0	0	Tetrahedral	Tetrahedral
NH_3	3	0	1	"	Trigonal pyramidal
H_2O	2	0	2	"	V shape or bent
$\text{CH}_2=\text{CH}_2$	3	1	0	Trigonal planar	Trigonal planar
$\text{CH}=\text{CH}$	2	2	0	Linear	Linear

Conjugated dienes :- A single bond separated by two double bond (or 4 sp^2 carbon in a row)



3. Cumulated Dienes Both sets of double bond are located on the same carbon atom. $\text{CH}_3 - \text{CH} = \text{C} = \text{CH} - \text{CH}_3$

4. Stability Dienes :- the relative stabilities of dienes are arranged in the following order: $\text{Isolated} > \text{conjugated} > \text{cumulated}$

Commonly occurring form of Carbon :- Carbon's most common form are :-

- ① diamond ② Graphite ③ Carbides
④ Fullerenes ⑤ charcoal.

NOTE :- Each C in a diamond is sp^3 the solid is tetrahedral.

* Each C in Graphite is sp^2 solid layers with weak van der waal forces b/w them.

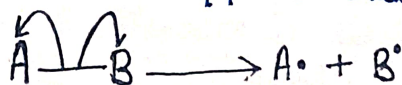
Breaking of Bond :-

The bond that is important for the study of rxn in organic chem. is the covalent bond. As a result we investigate how a covalent bond can be broken.

- ① Homolytic fission ② fission or Heterolytic

Homolytic fission / Homolytic cleavage

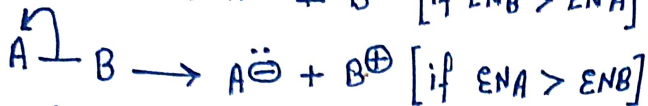
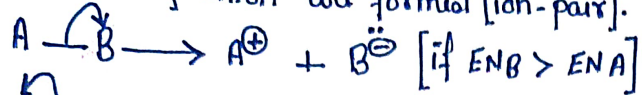
Each atom separate with one electron in this type of bond breaking resulting in the formation of highly reactive species k/w as radicals [free radicals].



Two half headed or fishhook represent bond breaking. One electron movement is depicted by a half headed arrow. Radicals are odd e^- species that are neutral

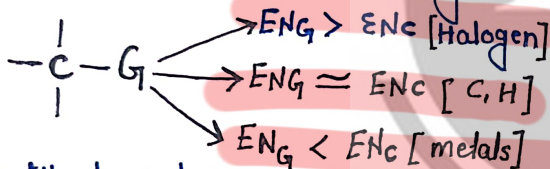
Heterolytic fission or Homolytic cleavage:

The shared pair of e^- are transferred to the more electronegative part in this type of covalent bond breaking. As a result of this fission a cation & anion are formed [ion-pair].



$\therefore$ A full headed arrow represent the bond break. A full head arrow depicts the movement of two e^- . In organic chemistry e^- movement is always represented by curved arrows either half headed or full headed.

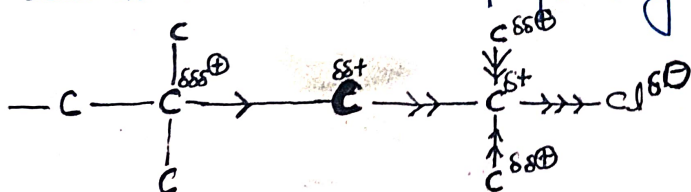
INDUCTIVE Effect:- When two unlike atom form a covalent bond the electron pair that form the sigma bond is never shared equally b/w two atom but is shifted slightly toward the more electronegative species.



As illustrated there are three type of grp/atom that can be attached to carbon. Although C is more electronegative than H, the difference in electronegativity is small & the bond is generally considered nonpolar.

Nature of Inductive Effect:- The Inductive effect is permanent effect that is the directly $\propto$ to dipole movement. It is minor effect because e^- are only shifted via sigma bond.

Effect of Branched Carbon chain:- An ex of operation of the inductive effect has been marked which is self explanatory.



Electron donating & Electron withdrawing group:- The Ind. effect can be caused by a single atom or by a grp of atom. The relative I.E. are measured by in relation to hydrogen. Electron donating grp or Electron releasing grp are those that donate electron to the carbon chain & are said to exert a +I effect. EWG are those that withdraw electron from the carbon chain and are said exert the -I effect.

Important:-

1. I.E. of alkyl grp $3^\circ > 2^\circ > 1^\circ > 0$
2. In general the no. of C in alkyl grp $\uparrow$ the +I effect.
3. To solve the problem we assume that the EN of sp hybridized C is greater than that sp^3 hybridized nitrogen.

Application of Inductive effect

Effect On Acidic / Basic strength:-

EWG raises acidic strength while $\downarrow$ basic strength. $\star$ ERG reduces acidic strength while increasing basic strength.

Ex:- **compare acidic strength**:-
 (i) $\text{CH}_3\text{---COOH}$ (ii) $\text{CH}_2\text{Cl---COOH}$ (iii) $\text{CH}_2\text{F---COOH}$

Solⁿ:-

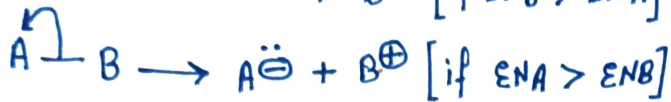
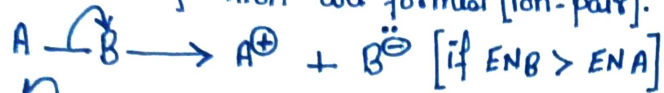
An alkyl grp will only donate if no other EWG is att . As a result EWG such as $\text{---CH}_2\text{Cl}$ & CH_2F form.

(iii) > (ii) > (i) $\rightarrow$ acidic strength.

Effect of Distance:- The I effect decrease as the ERG/EWG moves away.

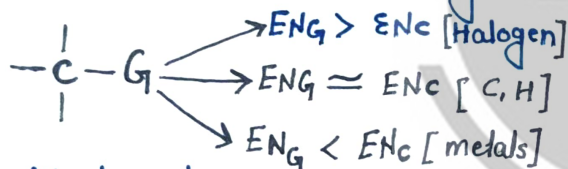
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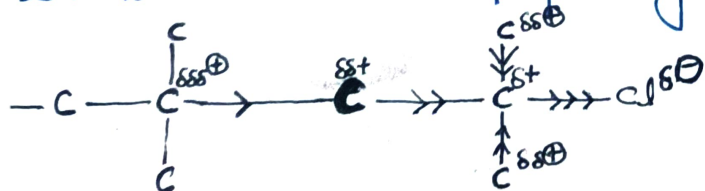
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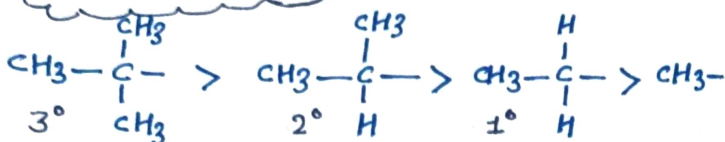
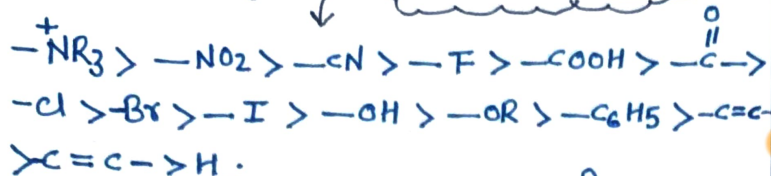
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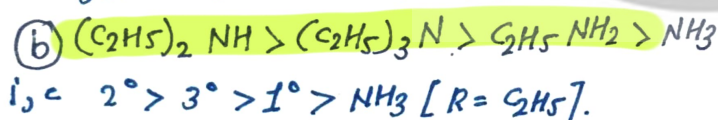
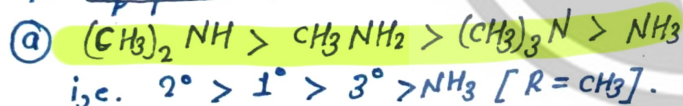
Series of +I and -I group in order of their strength :-



→ D - > H

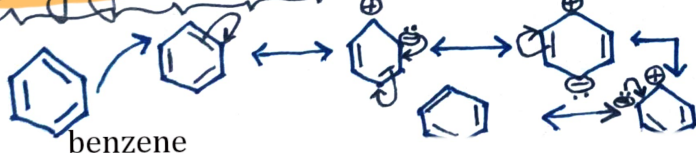
Basicity of Amines :- The order in gaseous or liq. phase is 3° > 2° > 1° > NH₃.

To ascertain the basic strength of amine in aq. solⁿ. We must consider the I effect, the solvation effect & the steric hindrance. The order of basic strength is thus experimental in the aq. state because we cannot prioritize the stability provided by any single effect. For the aq phase two result are critical:

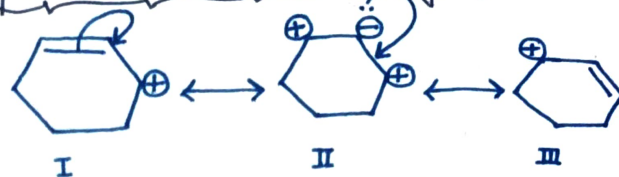


RESONANCE :- Simple Lewis structure are commonly used to represent molecule but some molecule can't be represented by single Lewis struct. As a result the concept of resonance was discovered. The delocali. of e⁻ [generally pi e⁻] is referred to resonance.

Conjugated System :- pi alternate pi



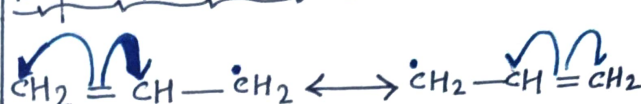
pi alternate positive charge :-



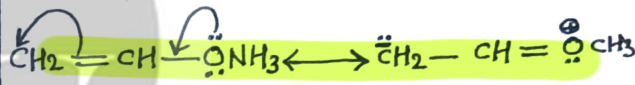
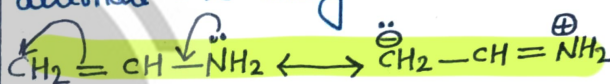
pi Alternate Negative charge :-



pi Alternate Odd electron :-



pi Alternate Lonepair :- Since lonepair and -ve charge are treated similarly, this case is similar to pi alternate -ve charge.



Lonepair and positive charge on Adjacent atom



Rule for Validity of Lewis structure :-

Rule 1. All Lewis str. must adhere to the Lewis Octet rule.

Rule 2. Atom in all resonating str. must be in the same position only the electron are in motion.

Rule 3. All resonating str. must have the same no. of paired & unpaired e⁻ that is the sum of bond pair and lonepair must be constant.

Rule 4. All of the atom in a molecule that participate in resonance must be coplanar. This is req. for effective p orbital overlap & e⁻ delocalised as in

in buta 1,3 diene.

Criteria for major/minor Contributors :-
Can be used to compare resonance form.

1. Use as many octet as you can [a neutral molecule always more stable in which its octet is complete].
2. Attempt to form as many bond as possible.
3. $+ve$ of a $\ominus$ charge on a more EN atom is stable.
4. The separation of charges.

(a) Similar charges :- keep them as far away from each other as possible to reduced repulsion & instability.

(b) Opposite charges :- should be kept as close together as possible to maximize attraction & stability.

Eg:- which of the str. is more stable.

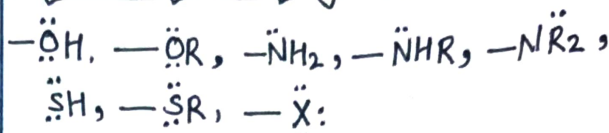


∴ II is more stable as all the octet are complete.

MESOMERIC EFFECT :- The permanent polarisation caused by a grp conjugated with a pi bond or a set of alternate bond is transmitted through the system electron resulting in a diff e^- distribution in the unsaturated chain. Mesomeric effect or resonance effect refer to the redistribution of electron in unsaturated compound conjugated with e^- releasing or e^- drawing grp. the dipole moment indicates that this effect is permanent.

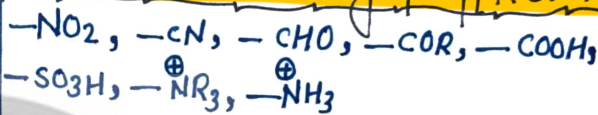
Electron-Releasing & Electron Withdrawal Grp :- the M or R effect refer to grp that release or withdraw e^- via resonance.

Electron-Releasing Group [R / +M eff]



The atom connected with the conjugated system has a lone pair to donate which is shared by all of the grp listed. As a result a generic representation can be represented. $-\ddot{\text{Z}}$

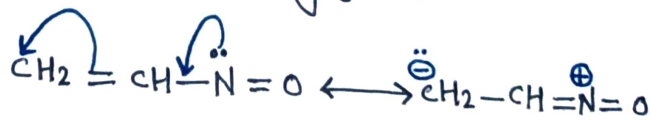
Electron-Withdrawing Group [R or -M eff]



∴ All of the grp listed have one thing in common: the atom connected with the conjugated system has a bond with another more EN atom that either withdraws e^- or directly has a positive charge on them. As a result a generic representation can be represented $\text{Y} = \text{Z}$ [$\text{EN}_\text{Z} > \text{EN}_\text{Y}$]

Dual Behaviour :- As shown some grp are both e^- releasing & e^- withdrawing

As electron releasing grp :-



As electron withdrawing grp :-



which behaviour dominate & which is use in a particular context will be discussed later in Electrophilic Aromatic substitution later. Resonance effect does not depend upon distance unlike inductive effect.

Application of Mesomeric effect:

Effect on Acidic strength of Carboxylic acid

& phenol: The resonating struct. of carboxylic acid result in a charge-separated struct. that is less stable than the charge delocalised carboxylate ion. As a result carboxylic acid easily loses proton (H) to form a carboxylate ion.

∴ Similarly in PHENOL resonance cause charge separation, which increase the rate of ionization and result in the formation of phenoxide ion, which is stabilized by charge delocalization.

Effect On Reactivity of Carboxylic Acid derivatives

∴ A typical nucleophilic rxn is represented as: The stronger is the bond b/w C & Z the more difficult it is for a nucleophile to break a bond & therefore, lower reactivity. Reactivity order carboxylic acid derivative towards nucleophilic acyl substitution is

Acyl chloride > Acid Anhydride > Ester
> Amide

Effect of ERG | EWG on acidic/

Basic strength ∴ EWG increase
the acidic strength &
decrease the basic strength

ERG decrease the acidic strength &
increase the Basic strength